Differentiable Manifolds

Nishtha Tikalal

1 Motivation

A topological manifold $(M, \mathcal{O})$ lets us talk about continuity of curves: a particle's trajectory does not jump. We now also want to associate a *velocity* to a curve at each point, which requires a unique tangent vector, i.e. *differentiability*. A curve with a kink has no unique tangent there, whereas a smooth one does.

The structure $(M, \mathcal{O})$ is *not* enough to define differentiability of curves, of functions $M \to \mathbb{R}$ (e.g. a temperature distribution), or of maps between manifolds. We must make additional choices. This lecture determines what those choices are and how many there are. Preview of the answer:

- in dimension $1, 2, 3$ there is essentially a single choice;
- in dimension ≥ 5 there are finitely many;
- in dimension 4 (spacetime!) there can be uncountably many.

This is not mathematical sophistication for its own sake: it matters for physics which structure we choose.

2 The strategy and why it fails at first

Strategy. Let $\gamma : \mathbb{R} \to M$ be a curve. Choose a chart (U, x) and consider the portion of γ in U. Then

$$x \circ \gamma : \mathbb{R} \to x(U) \subseteq \mathbb{R}^d$$

is a curve in $\mathbb{R}^d$, for which we know what differentiability means (undergraduate differentiability). Try to *lift* this to a notion of differentiability ("manifold differentiability") of γ on M. The same idea would apply to functions $f : M \to \mathbb{R}$ (use $f \circ x^{-1}$) and to maps $\phi : M \to N$ (use charts on both sides).

Well-definedness. The curve knows nothing about our choice of chart, so the definition must be independent of it. Take another chart (V, y) with $U \cap V \neq \emptyset$ and consider the part of γ in the overlap. Then

$$y \circ \gamma = (y \circ x^{-1}) \circ (x \circ \gamma),$$

where $x \circ \gamma$ is differentiable by assumption and $y \circ x^{-1}$ is the chart transition map (a map $\mathbb{R}^d \supseteq \cdot \to \mathbb{R}^d$). For a topological manifold the transition map is only guaranteed to be *continuous*. The composition of a differentiable map with a merely continuous one may fail to be differentiable (a continuous transition map can introduce an edge). So $y \circ \gamma$ may be non-differentiable even though $x \circ \gamma$ is, and the strategy fails at first sight.

3 Compatible charts

So far we used arbitrary charts from the *maximal atlas* $\mathcal{A}_{\max}$ of $(M, \mathcal{O})$: the atlas containing every possible chart. The remedy is to throw out charts whose transition maps are not good enough, keeping a sub-atlas that is still an atlas (still covers M) but whose transition maps all have the required property. Then we only define differentiability using charts from the restricted atlas.

Definition 1 (Compatible charts)**.** Let (U, x) and (V, y) be charts of a topological manifold, and let $*$ be a property of maps $\mathbb{R}^d \to \mathbb{R}^d$ (e.g. C^k). The charts are called $*$*-compatible* if either

- $U \cap V = \emptyset$ (charts that never meet cannot be incompatible), or
- $U \cap V \neq \emptyset$ and both chart transition maps

$$y \circ x^{-1} : x(U \cap V) \to y(U \cap V), \qquad x \circ y^{-1} : y(U \cap V) \to x(U \cap V)$$

have property $*$.

Definition 2 ($*$-atlas and $*$-manifold)**.** An atlas $\mathcal{A}_* \subseteq \mathcal{A}_{\max}$ is a $*$*-atlas* if any two of its charts are $*$-compatible. A $*$*-manifold* is a triple $(M, \mathcal{O}, \mathcal{A}_*)$, where $(M, \mathcal{O})$ is a topological manifold and $\mathcal{A}_*$ is a $*$-atlas.

Choosing $\mathcal{A}_*$ is a *choice*, and it matters: for a given topological manifold there may be different, inequivalent sub-atlases with the same property, and with one choice certain curves are differentiable while with another they are not. (Tearing out particular pages of the maximal atlas also carries information.)

3.1 Examples of properties $*$

- C^0: every atlas of a topological manifold is a C^0 atlas, since transition maps are continuous (w.r.t. the standard topology on $\mathbb{R}^d$).
- C^k: transition maps are k times *continuously* differentiable. Requiring the derivatives to be continuous is preferable because C^k can then be checked via existence and continuity of the partial derivatives; there are maps all of whose partial derivatives exist but which are not nicely differentiable.
- C^∞ (smooth): arbitrarily often continuously differentiable.
- C^ω (real analytic): transition maps can be Taylor expanded (multidimensional Taylor expansion). This is a much smaller class than C^∞; there are smooth functions not analytic in any dimension. One needs this, e.g., to speak of Taylor-expandable curves.
- *Complex*: for even dimension $2d$, require that the transition maps $\mathbb{R}^{2d} \to \mathbb{R}^{2d}$ satisfy the Cauchy–Riemann equations. This gives a *complex manifold*, with much more structure.

The more structure we want on objects on the manifold, the more restrictive the atlas has to be.

3.2 Differentiability of curves

With a C^k atlas, a curve $\gamma : \mathbb{R} \to M$ is called C^k if $x \circ \gamma$ is C^k for charts (U, x) in the atlas. This is well defined: for another chart (V, y) of the atlas, $y \circ \gamma = (y \circ x^{-1}) \circ (x \circ \gamma)$ is a composition of C^k maps. (Analogously for functions and maps between manifolds.)

3.3 Whitney's theorem

Theorem 1 (Whitney, quoted without proof)**.** *Every C^k atlas with $k \geq 1$ of a topological manifold contains a sub-atlas (which is still an atlas) that is a C^∞ atlas.*

Some topological manifolds admit no C^1 atlas at all, so the step from C^0 to C^1 is the substantial one. But once a C^k atlas with $k \geq 1$ exists, one can trickle down to C^∞ by removing charts. Hence without loss of generality we always consider C^∞ manifolds, called **smooth manifolds**. (This does not extend to the analytic or complex cases: an underlying topological manifold may not admit further reduction of the atlas.)

Definition 3 (Smooth manifold)**.** A *smooth manifold* is a triple $(M, \mathcal{O}, \mathcal{A})$ where $(M, \mathcal{O})$ is a topological manifold and $\mathcal{A}$ is a C^∞ atlas.

Remark (Curves need not be C^∞)*.* Choosing a smooth manifold does not force every curve to be smooth. One can still consider curves that are only C^2 (say, if the fifth derivative of a trajectory is discontinuous, as for a hammered nail): a C^∞ transition map composed with a C^2 curve in a chart gives a C^2 curve in the other chart. Whitney's theorem only says it does not harm C^k objects to choose the manifold C^∞.

Remark (Real world versus chart)*.* When a trajectory is written down as a function in some chart, we are talking about $x \circ \gamma$; the physical object is γ itself (the real path of the real chalk). Differentiability of γ is meaningful only because we restricted to a C^∞ atlas.

4 Diffeomorphisms

As usual, structure is studied via structure-preserving maps.

Sets. For "naked" sets M, N the structure-preserving maps are *bijections*. M and N are *set-theoretically isomorphic* if a bijection $\phi : M \to N$ exists. Examples: $\mathbb{Z} \cong \mathbb{N}$ and $\mathbb{Q} \cong \mathbb{N}$ (set-theoretically isomorphic), but $\mathbb{N}$ is not set-theoretically isomorphic to $\mathbb{R}$; $\{1, 2, 3\}$ and $\{A, B\}$ are not isomorphic.

Topological spaces. Two topological spaces are *homeomorphic* (topologically isomorphic) if there is a bijection ϕ with ϕ and ϕ^{-1} both continuous. Homeomorphic implies set-theoretically isomorphic, but not conversely.

Vector spaces. Two vector spaces are isomorphic if there is a bijection that is linear (in both directions).

4.1 Smooth maps and diffeomorphisms

Definition 4 (Smooth map)**.** Let $(M, \mathcal{O}_M, \mathcal{A}_M)$ and $(N, \mathcal{O}_N, \mathcal{A}_N)$ be smooth manifolds of dimensions d and e. A map $\phi : M \to N$ is C^∞ if for all charts $(U, x) \in \mathcal{A}_M$ and $(V, y) \in \mathcal{A}_N$ the map

$$y \circ \phi \circ x^{-1} : \mathbb{R}^d \supseteq x(U \cap \phi^{-1}(V)) \to y(V) \subseteq \mathbb{R}^e$$

is C^∞ in the undergraduate sense. This does not depend on the charts used: other charts $\tilde{x}, \tilde{y}$ differ by chart transition maps, which are C^∞ because the atlases are C^∞.

Definition 5 (Diffeomorphic)**.** Two smooth manifolds $(M, \mathcal{O}_M, \mathcal{A}_M)$ and $(N, \mathcal{O}_N, \mathcal{A}_N)$ are *diffeomorphic* if there exists a bijection $\phi : M \to N$ such that both ϕ and ϕ^{-1} are C^∞ maps. Such a ϕ is called a *diffeomorphism*.

Example 1.

- The unit sphere and an ellipsoid, both in $\mathbb{R}^3$ with the induced topology and a smooth atlas, are diffeomorphic.
- A sphere and a sphere with a small edge or fold put into it are homeomorphic, hence the same topological manifold, but are not diffeomorphic as smooth manifolds.
- The surface of a potato without folds is diffeomorphic to the surface of the Earth. At the level of smooth manifolds, objects do not yet have a "shape"; that comes later, with more structure.

5 How many smooth structures?

Counting smooth manifolds that can be made out of a given topological manifold (if any exists), up to diffeomorphism, as a function of its dimension d:

d	number of smooth structures up to diffeomorphism
$1, 2, 3$	exactly one (Moise–Radó; transcribed in the lecture as "Moyes Radon")
4	generally **uncountably infinitely many**, e.g. for non-compact manifolds; only special partial result
≥ 5	finitely many (via surgery theory); the number may depend on the manifold

Surgery theory required new techniques in topology (gluing manifolds together etc.), so these are deep results. The lecturer is not aware of a definite number in the finite cases and expects it varies from manifold to manifold.

In dimension four, which is the dimension of spacetime, there is thus no single obvious smooth structure to choose, and one cannot simply "let experiment decide" among uncountably many options. This is a real problem for modeling spacetime.

Summary

- A topological manifold does not suffice to define differentiability: transition maps are only continuous, so chart-based differentiability is chart-dependent.
- Fix by restricting to a sub-atlas of mutually C^k-compatible charts; this is extra structure (a choice).
- By Whitney's theorem we may take C^∞ atlases: a **smooth manifold** is a triple $(M, \mathcal{O}, \mathcal{A})$ with $\mathcal{A}$ a C^∞ atlas. Analytic and complex structures do not follow automatically.
- Smooth maps are defined through charts; structure-preserving bijections of smooth manifolds are **diffeomorphisms**.
- Smooth structures on a topological manifold, up to diffeomorphism: unique in $d \leq 3$, finitely many in $d \geq 5$, possibly uncountably many in $d = 4$.